

Pre-Workshop Instructions

1. Complete sign-in form using the QR code on your table
2. Navigate to github.com/pyiu-workshops/pyiu
3. Click on “**crash-course**”, then “**artemis-ii**”
4. Click on “**artemis_ii_demo.ipynb**”, click on the three dots in the top right corner, and click “Download”
5. Upload this file to your IU Google Drive, and double-click it. This will open Colab!
6. You’re now ready to code with us!



PyIU Python Crash Course

AN INTRODUCTION TO PYTHON FOR STUDENTS
INTERESTED IN ASTRONOMY/STEM

By: IU Astronomy & Physics Graduate Students

09/16/26



Crash Course Purpose

- Astronomers, physicists, and other scientists use **programming** as a tool to analyze data for research
- Python has become the most-used programming language, becoming a vital skill!

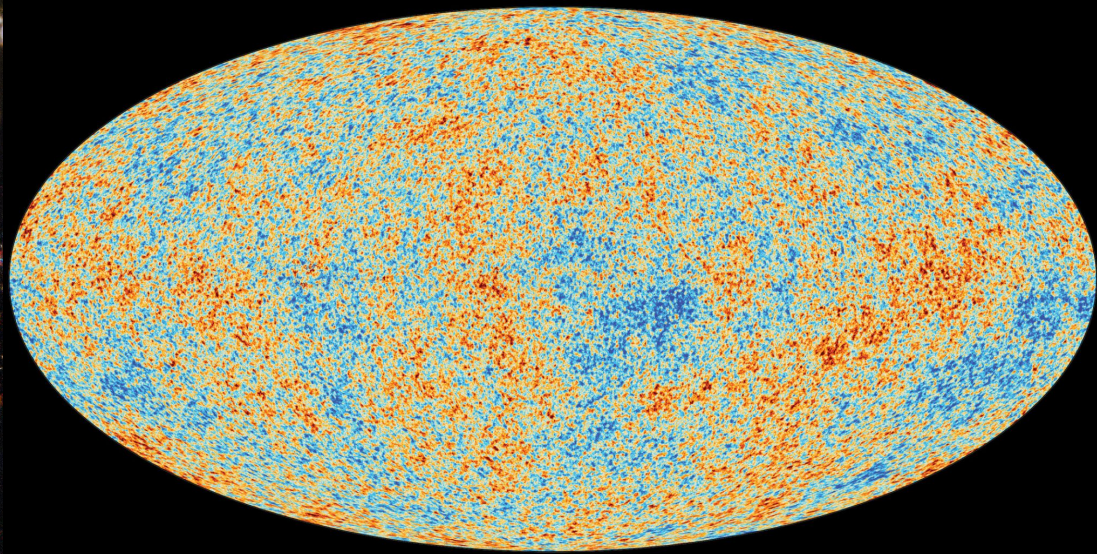


But Python is much more user-friendly than these!
(at the cost of being a bit slower)

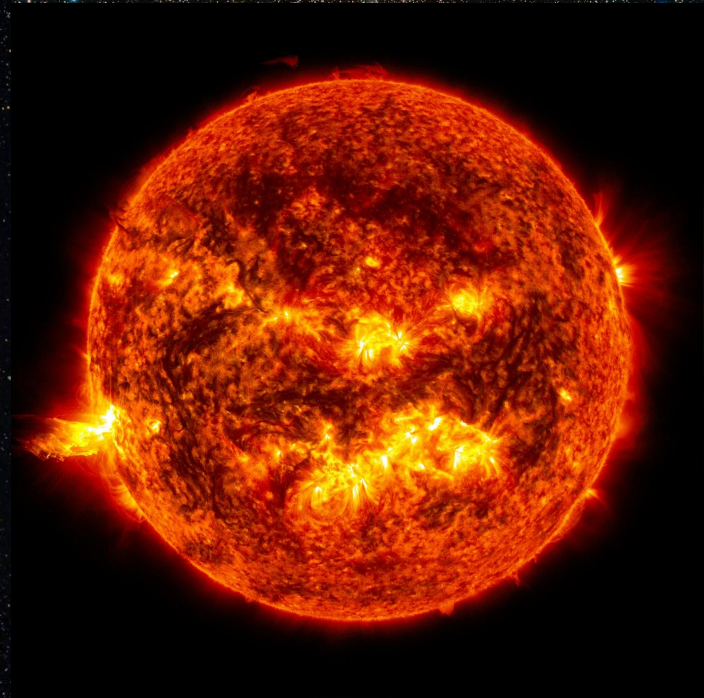
Other common languages:

C, C++, C#
Java & JavaScript
SQL
R
MATLAB
Rust
Perl
HTML & CSS
Fortran
Julia

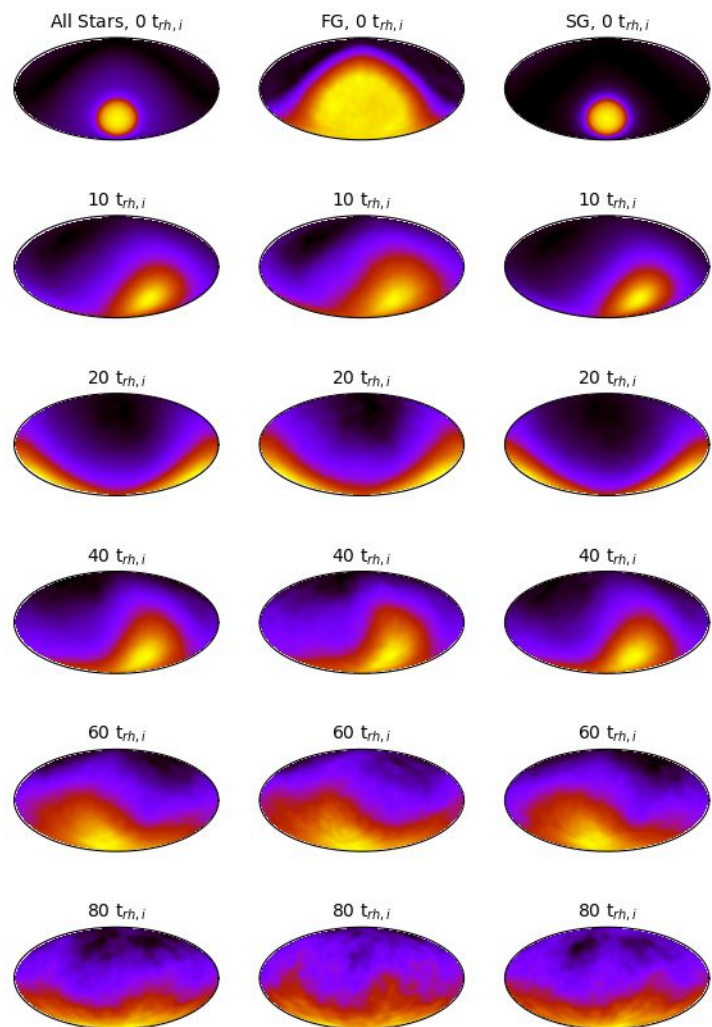




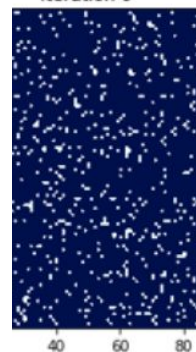
Coding is your gateway to the universe...



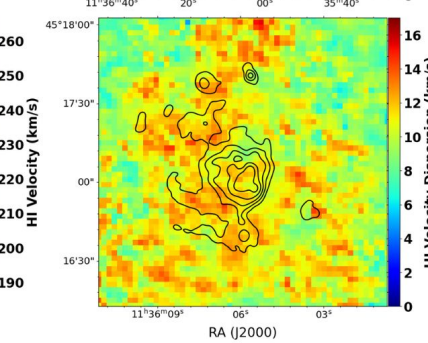
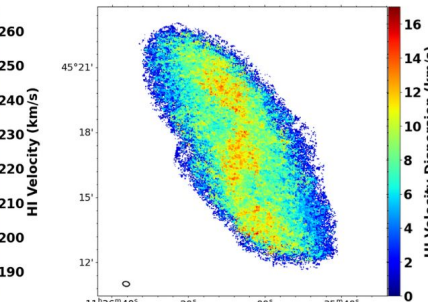
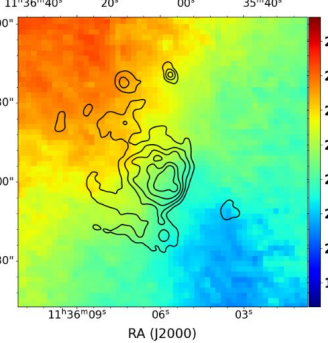
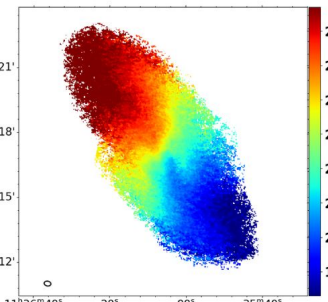
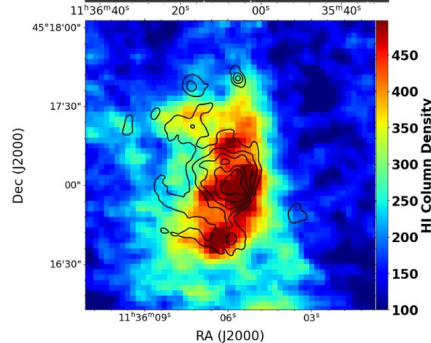
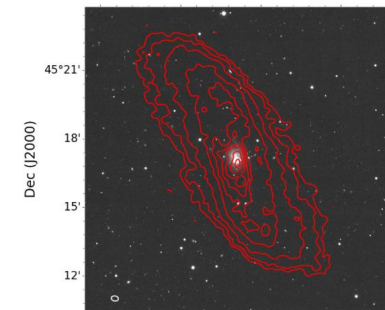
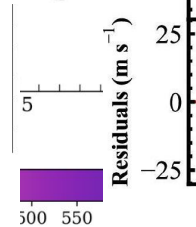
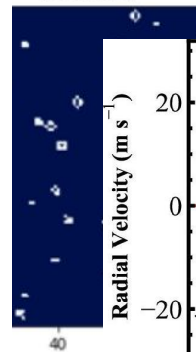
What we've done with Python!



Iteration 0



Iteration 4



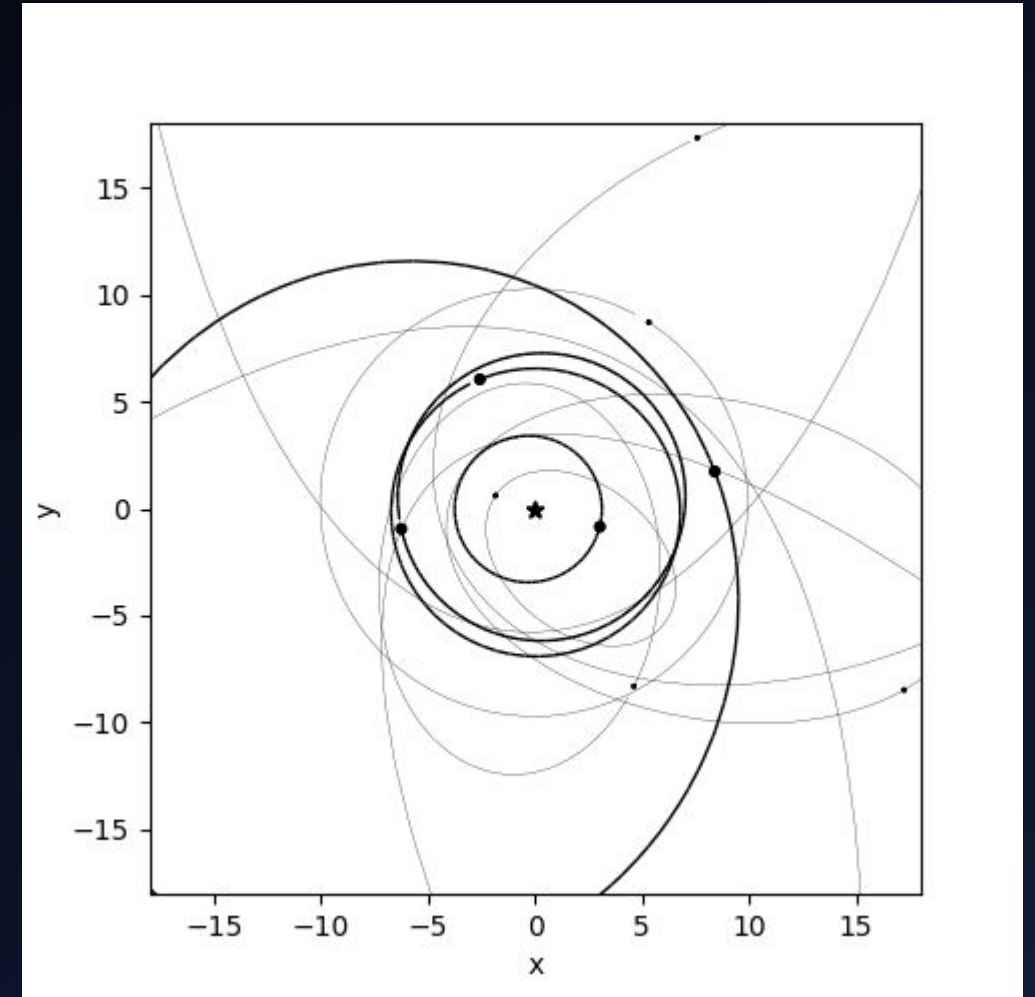
Radial Velocity (m s^{-1})

Residuals (m s^{-1})

Time from Mid-Transit (hours)

Time from Mid-Transit (hours)

More Examples of Python from IU Astronomy



Python



Pros of Python:

- Simple and user-friendly
- Interpreted, high-level language

Documentation:

<https://docs.python.org/3/>

Cons:

- A bit of a black box, slow
- Interpreted, high-level language

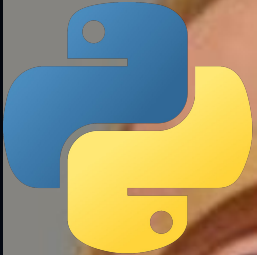
Lots of “layers” to executing the code, computes line-by-line

Easily readable to humans... not so much for a computer



A Demonstrative Example: “Hello World”

Python



```
In [1]: ► print("Hello World!")  
Hello World!
```

C, C++, Fortran, IDL



```
#include <iostream>
```

```
int main() {
```

```
    std::cout << "Hello World!";
```

```
    return 0;
```

```
}
```

```
C:\Users\bradz>g++ -o hello helloworld.cpp
```

```
C:\Users\bradz>./hello
```

Where to write code in Python



Where to write code in Python



Where to write code in Python

- IDEs: Integrated Development Environments



PyCharm



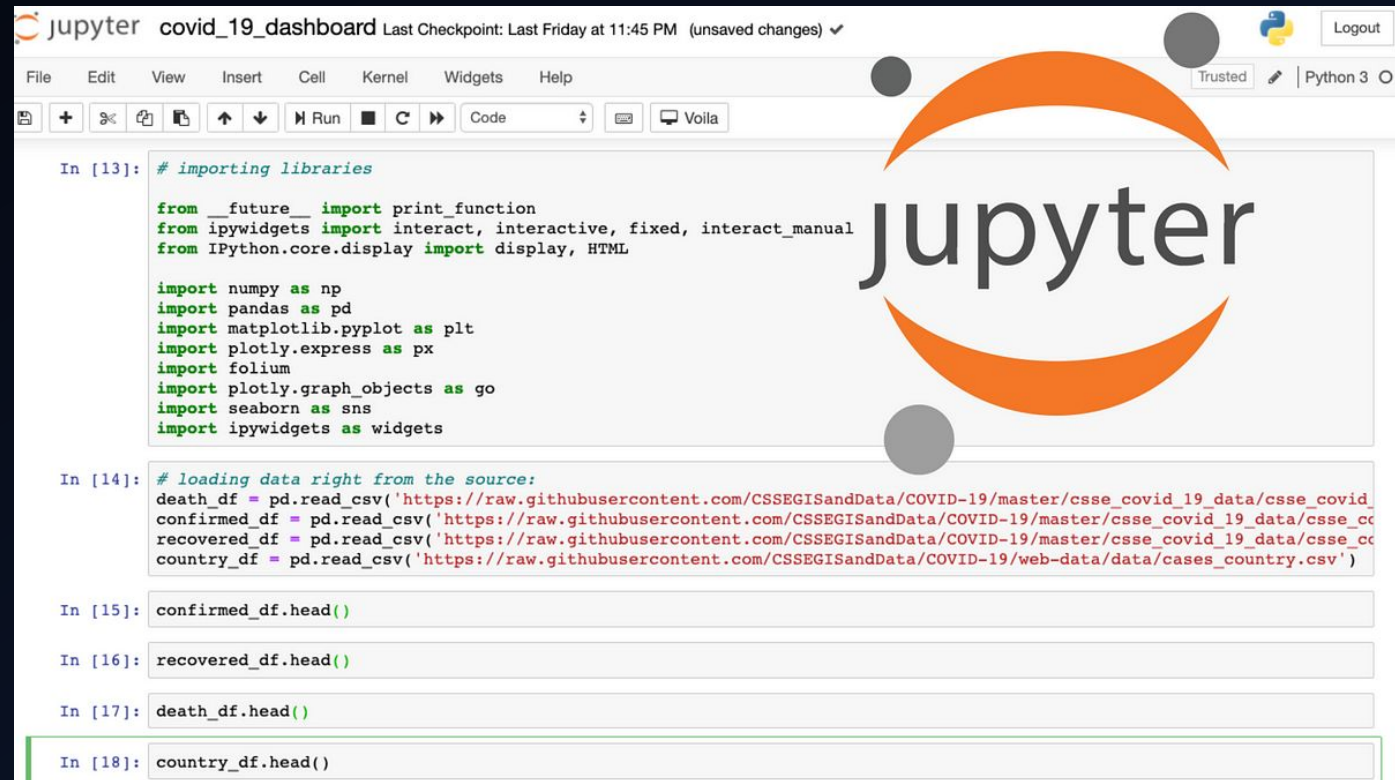
VSCode

What we're using for the Crash Course

• Jupyter notebook:
web-based interactive
Python development
tool

• Cell-by-cell code
blocks allow for:

- Easy debugging & visualization
- Quick and efficient code development



The screenshot shows a Jupyter Notebook window titled "covid_19_dashboard". The interface includes a top menu bar with options like File, Edit, View, Insert, Cell, Kernel, Widgets, and Help. Below the menu is a toolbar with icons for running code, saving, and other actions. The main area displays a series of code cells. The first cell (In [13]) imports various libraries including numpy, pandas, matplotlib, plotly, folium, seaborn, and ipywidgets. The second cell (In [14]) loads COVID-19 data from a GitHub repository using pandas' read_csv function. Subsequent cells (In [15] to In [18]) show the head of the loaded dataframes (confirmed_df, recovered_df, death_df, and country_df). A large Jupyter logo is overlaid on the right side of the notebook interface.

```
In [13]: # importing libraries
from __future__ import print_function
from ipywidgets import interact, interactive, fixed, interact_manual
from IPython.core.display import display, HTML

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import plotly.express as px
import folium
import plotly.graph_objects as go
import seaborn as sns
import ipywidgets as widgets

In [14]: # loading data right from the source:
death_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_data/death_df.csv')
confirmed_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_data/confirmed_df.csv')
recovered_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_data/recovered_df.csv')
country_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/web-data/data/cases_country.csv')

In [15]: confirmed_df.head()

In [16]: recovered_df.head()

In [17]: death_df.head()

In [18]: country_df.head()
```

The Artemis II Mission

PyIU Python Crash Course



The Goal

- Establish a permanent human presence on the Moon
- Prepare for a crewed mission to Mars
- Artemis II was a test of NASA's capability to send a crewed vessel to the Moon and back



Image Credit: NASA

The Astronauts

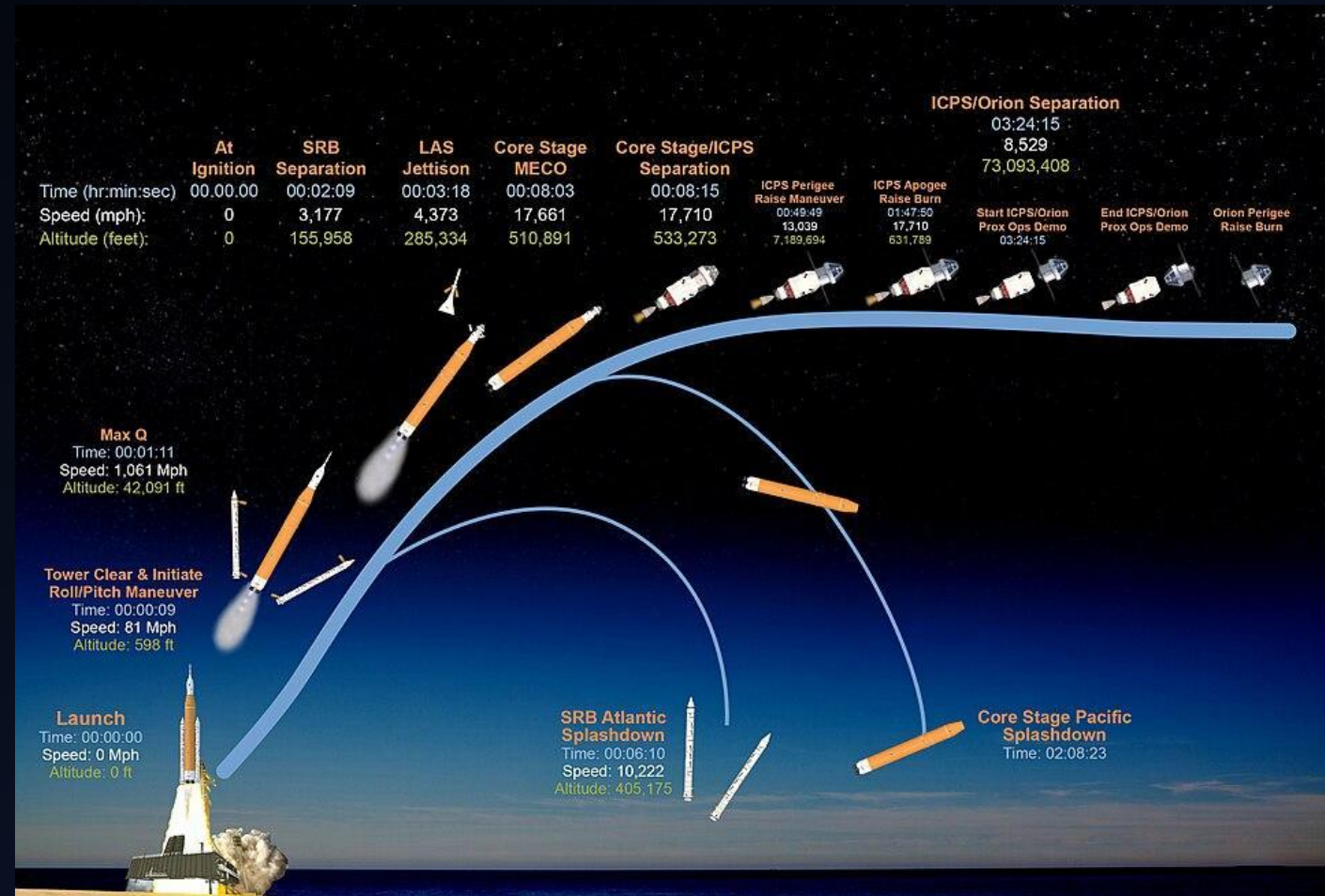
- Christina Koch (NASA): Mission Specialist 1
- Victor Glover (NASA): Pilot
- Jeremy Hansen (CSA): Mission Specialist 2
- Reid Wiseman (NASA): Commander



Image Credit: NASA

The Launch

- Launched from Kennedy Space Center in Florida on April 1, 2026, at 6:35 pm EDT
- [Launch video!](#)



The Journey

- Orion capsule flew around the Moon and back in 9 days, traveling around 700,000 miles (~1,100,000 km)
- Astronauts took lots of beautiful pictures throughout their journey



The Journey



Image Credit: NASA



The Journey



Image Credit: NASA



Your Turn

- Incrementally apply Python fundamentals
- Create your own visualization of the Artemis II mission's trajectory using real NASA data!



Image Credit: NASA



Pre-Workshop Instructions

1. Complete sign-in form using the QR code on your table
2. Navigate to github.com/pyiu-workshops/pyiu
3. Click on “**crash-course**”, then “**artemis-ii**”
4. Click on “**artemis_ii_demo.ipynb**”, click on the three dots in the top right corner, and click “Download”
5. Upload this file to your IU Google Drive, and double-click it. This will open Colab!
6. You’re now ready to code with us!



Let's Code!



Advanced Python Workshop: Coming this November!

- Learn techniques that you can apply directly to your research/upper-level coursework
- Casual, highly interactive environment with experienced instructors
- Potential topics:
 - Python installation/environments
 - Machine learning
 - Advanced plotting
 - Statistics
 - Numerical integration & simulations
 - Big data analysis & data fitting
 - YOU tell US!

